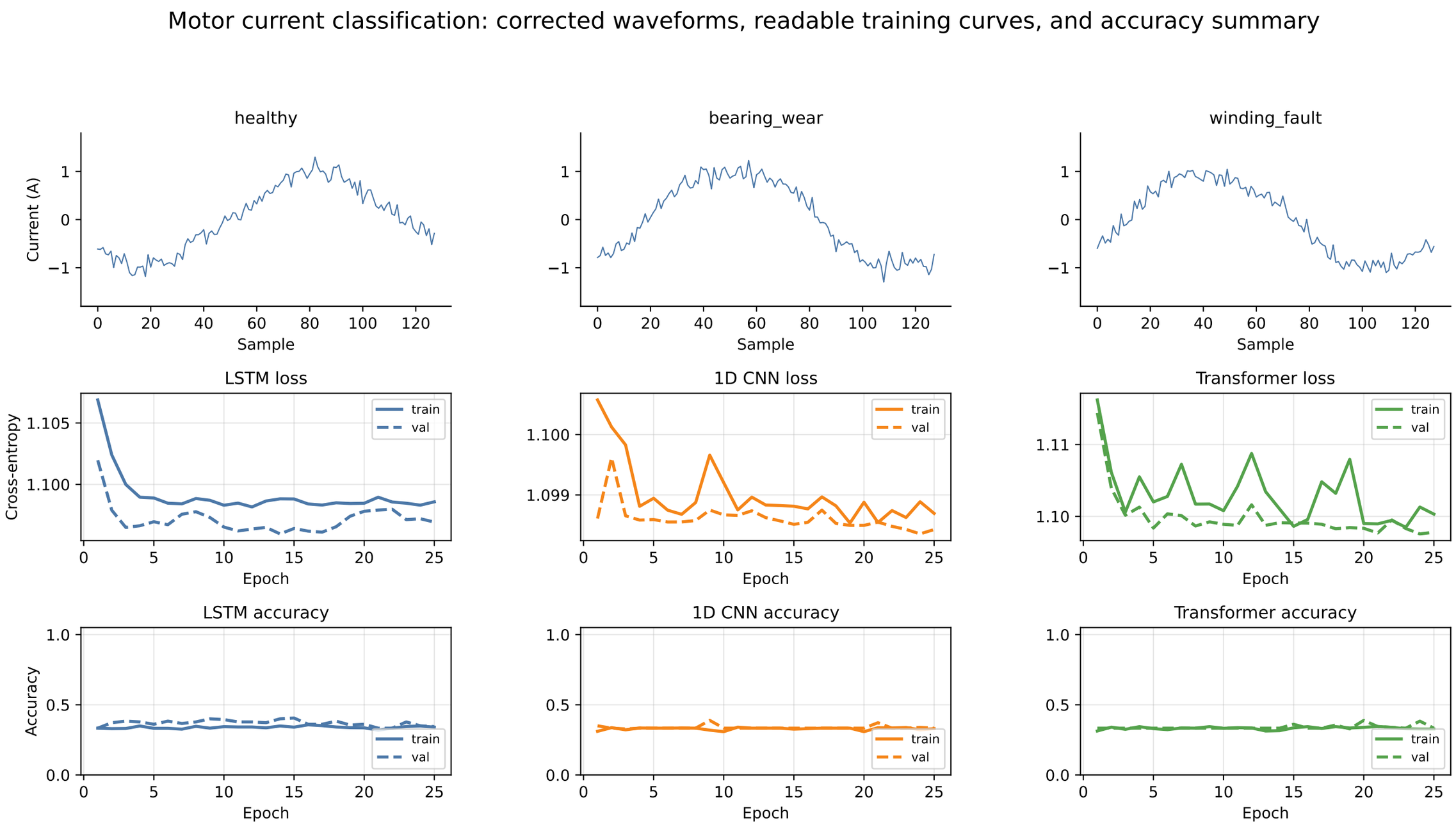
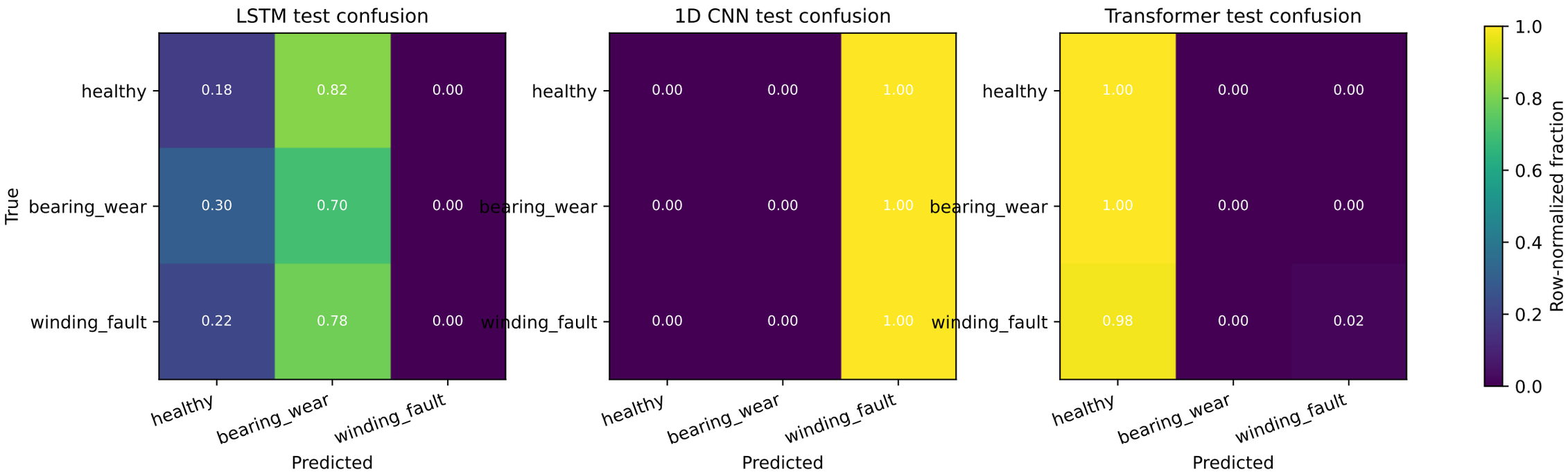


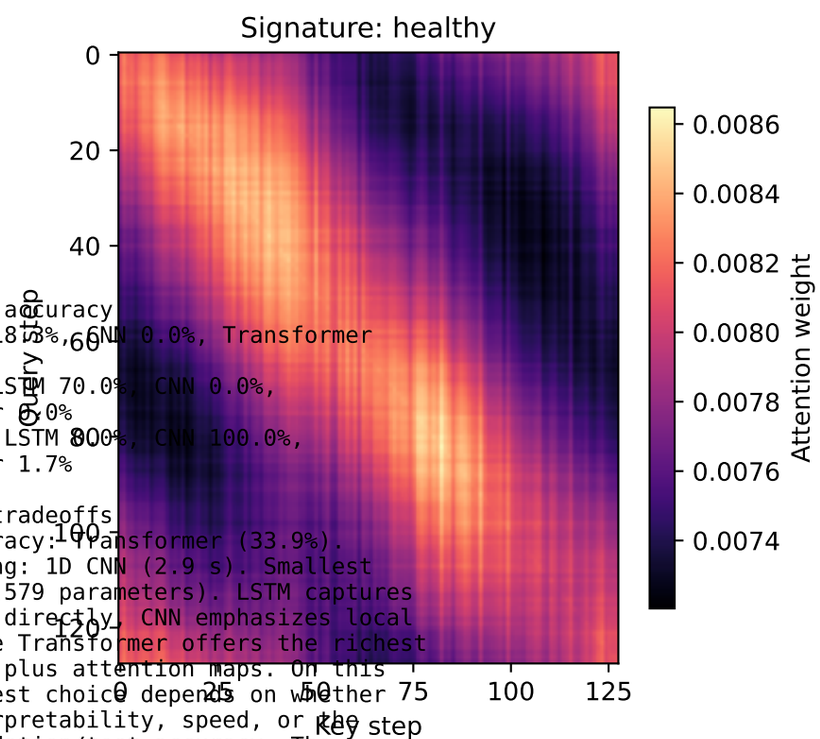
Best test accuracy: Transformer | Fastest: 1D CNN | Smallest: LSTM

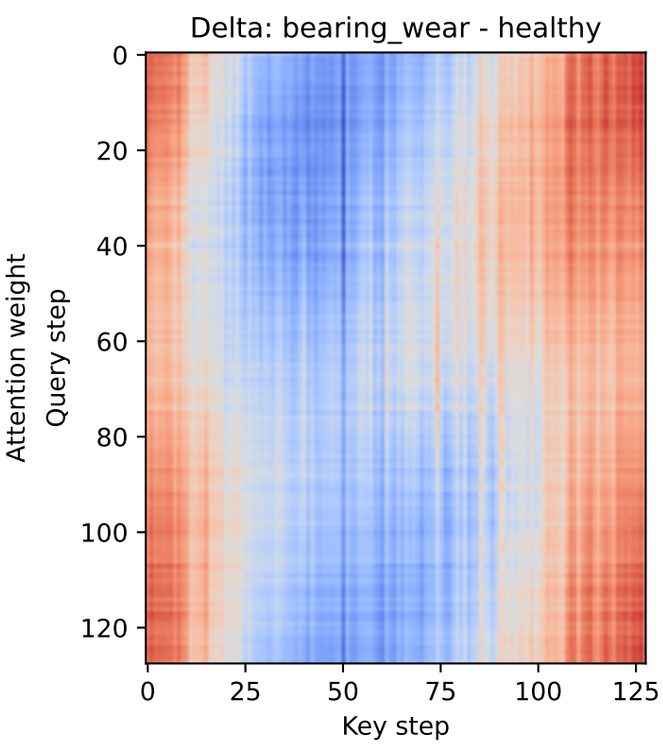
Accuracy comparison table summarizing the three architectures on the held-out test set.

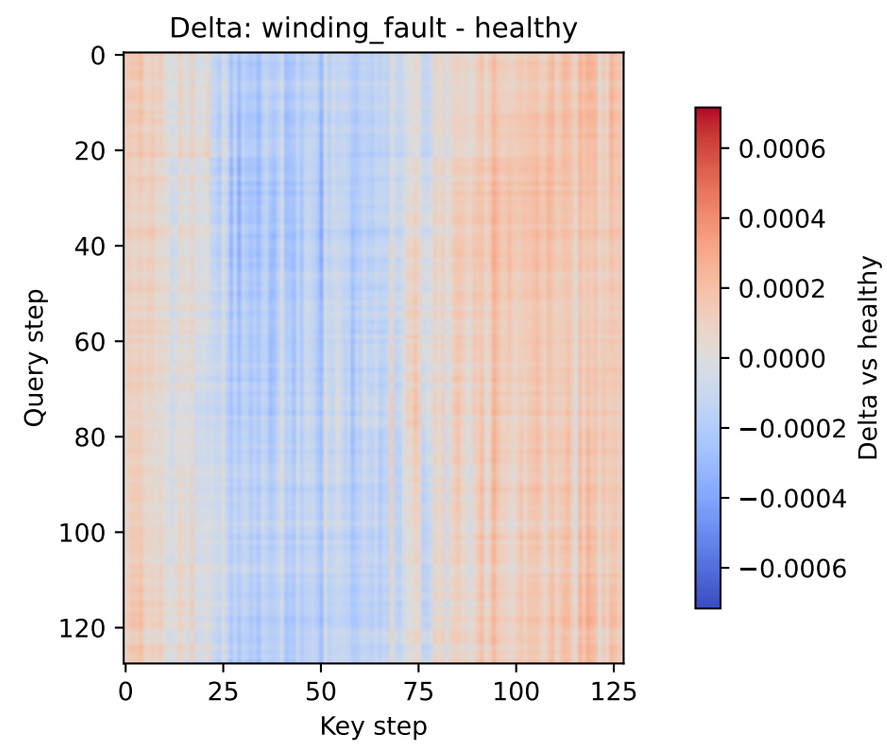
Model	Params	Test acc	Final val acc	Train time
LSTM	4,579	29.4%	34.4%	8.8 s
1D CNN	13,187	33.3%	33.3%	2.9 s
Transformer	21,347	33.9%	33.3%	12.2 s

Per-class behavior, confusion heatmaps, and class-signature attention deltas









Per-class test accuracy
healthy: LSTM 18.3%, CNN 0.0%, Transformer 100.0%
bearing_wear: LSTM 70.0%, CNN 0.0%, Transformer 0.0%
winding_fault: LSTM 80.0%, CNN 100.0%, Transformer 1.7%

Architectural tradeoffs
Best test accuracy: Transformer (33.9%). Fastest training: 1D CNN (2.9 s). Smallest model: LSTM (4,579 parameters). LSTM captures temporal order directly, CNN emphasizes local motifs, and the Transformer offers the richest global context plus attention maps. On this dataset, the best choice depends on whether you value interpretability, speed, or the strongest validation/test accuracy. The Transformer is the best candidate when attention visualizations matter and a small accuracy gain justifies extra complexity.